about what happens next: it quickly spreads out all over the place. Left to their own devices, Schrödinger's wave functions don't look particle-like at all.*

It was left to Max Born, one of Heisenberg's collaborators on matrix mechanics, to provide the final missing piece: we should think about the wave function as a way of calculating the probability of seeing a particle in any given position when we look for it. In particular, we should take both the real and imaginary parts of the complex-valued amplitude, square them both individually, and add the two numbers together. The result is the probability of observing the corresponding outcome. (The suggestion that it's the amplitude squared, rather than the amplitude itself, appears in a footnote added at the last minute to Born's 1926 paper.) And after we observe it, the wave function collapses to be localized at the place where we saw the particle.

You know who didn't like the probability interpretation of the Schrödinger equation? Schrödinger himself. His goal, like Einstein's, was to provide a definite mechanistic underpinning for quantum phenomena, not just to create a tool that could be used to calculate probabilities. "I don't like it, and I'm sorry I ever had anything to do with it," he later groused. The point of the famous Schrödinger's Cat thought experiment, in which the wave function of a cat evolves (via the Schrödinger equation) into a superposition of "alive" and "dead," was not to make people say, "Wow, quantum mechanics is really mysterious." It was to make people say, "Wow, this can't possibly be correct." But to the best of our current knowledge, it is.

A lot of intellectual action was packed into the first three decades of the twentieth century. Over the course of the 1800s, physicists had put together a promising picture of the nature of matter and forces. Matter was made of particles, and forces were carried by fields, all under the umbrella of classical mechanics. But confrontation with experimental data forced them to think beyond this paradigm. In order to explain radiation from hot objects, Planck suggested that light was emitted in discrete amounts of energy, and Einstein pushed this further by suggesting that light actually